Intrinsic Dimensionality and Geometric Structure in Pan-Cancer RNA-Seq Data

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Abstract

Tumor RNA-Seq profiling measures over 20,000 genes per patient while cohorts remain in the hundreds – a classic small-n, large-d regime in which standard statistical intuitions break down. Using the TCGA Pan-Cancer HiSeq dataset ($n = 801$ samples, $d = 20{,}531$ genes, 5 cancer types), we measure the gap between ambient and intrinsic dimensionality and test whether classical high-dimensional phenomena predicted by theory actually show up in this biologically structured data. After a justified preprocessing pipeline (log transform, zero-variance-gene removal, standardisation), three independent estimators are calibrated against synthetic Gaussian-noise and known-5-D-manifold baselines, with all three datasets run through one identical preprocessing path. The estimators are the Grassberger–Procaccia correlation dimension ($D_2 = 7.73$), the Levina–Bickel k-NN MLE (mean 29.36), and PCA variance/elbow criteria (100–392 components). We bring these together into a single account: the data occupy a low-dimensional manifold, but it is curved rather than flat. That curvature is why nonlinear embeddings (t-SNE, UMAP) separate the five types visually where linear PCA in the same number of components does not. A Johnson–Lindenstrauss random projection to 1,543 dimensions ($\varepsilon = 0.2$) preserves all pairwise distances within the theoretical bound. The empirical correlation-matrix spectrum reproduces the Marchenko–Pastur point mass exactly (60.00%), and it shows 27 outlier eigenvalues that reflect genuine covariance structure. Ledoit–Wolf/Mahalanobis anomaly detection flags zero outliers. We show that with $p = 2{,}000$ features and $n = 801$ samples, the mean squared distance is bounded above by $(n-1)/(1-\alpha)$ no matter what the data are. This means a χ^2_p threshold cannot be crossed by construction, and the rank of the estimator, not the feature count, sets the correct degrees of freedom. We close by discussing limitations, including small per-class samples (COAD, $n \approx 78$), unmodeled TCGA batch effects, and i.i.d. violations from gene co-expression, together with concrete next steps.

1 Introduction and Dataset

Modern tumor profiling technologies (RNA-Seq / Illumina HiSeq) routinely measure the expression of more than 20,000 genes per patient, while the number of patients in any given study is typically in the hundreds. This is the classic small-n, large-d regime. The ambient feature space is enormous, but biology suggests that tumor identity is governed by a much smaller number of underlying regulatory programs, such as pathways, co-expression modules, and cell-of-origin signals. With $d = 20{,}531 \gg n = 801$, standard statistical intuitions break down for density estimation, Euclidean nearest neighbors, and ordinary covariance estimation. Dedicated techniques are required to get reliable insight: dimensionality reduction, intrinsic-dimension estimators, regularised covariance estimation, and random-projection theory.

This project asks three questions. First, what is the *ambient* dimensionality of this dataset versus its *intrinsic* dimensionality, meaning the number of degrees of freedom actually needed to describe the variation between tumor samples? Second, do classical high-dimensional phenomena predicted by theory (concentration of measure, empirical spectral behavior of covariance matrices, distance preservation under random projection) actually show up in this real dataset, or does biological structure make it behave differently from random high-dimensional noise? Third, can clinically meaningful structure (five distinct cancer types) be recovered using only unsupervised, geometry-driven analysis of the raw feature space? The rest of this report answers these questions in three stages: preprocessing (Section 2), intrinsic-dimension estimation (Section 3), and geometric visualisation with random-matrix diagnostics (Section 4). Section 5 then ties the results together. Appendix A documents the AI tooling used to produce the code, figures, and drafts behind every result reported here.

1.1 Dataset and Provenance

The dataset is the Gene Expression Cancer RNA-Seq collection (TCGA Pan-Cancer Atlas, Illumina HiSeq platform): 801 tumor samples and 20,531 gene-expression features (`gene_0`...`gene_20530`), aspect ratio $d/n \approx 25.6$. Each sample carries one of five cancer-type labels: BRCA (breast), KIRC (kidney), LUAD (lung), PRAD (prostate), COAD (colon). All features are continuous, non-negative RNA-Seq intensities. This release (tcg) is a Kaggle mirror of the UCI ML Repository dataset (Fiorini, 2016, DOI 10.24432/C5R88H), itself sourced from the TCGA Pan-Cancer Atlas project (Weinstein et al., 2013), distributed under CC BY 4.0.

An integrity check (matrix shape, sample-ID alignment, missing-value and zero-variance counts) found no missing values and 267 of 20,531 genes ($\approx 1.3\%$) with exactly zero variance across all 801 samples. These are most likely genes that are simply not expressed, or not reliably measured, in these tissue types, rather than a data-quality problem.

Class balance is moderately imbalanced (Table 1, Figure 1): BRCA accounts for 37.5% of samples while COAD accounts for only 9.7%. This is a property of the raw data, and preprocessing does not change it, but it limits the statistical power of any class-conditional geometric analysis (Section 6).

Table 1: Class balance across the five cancer types.

Cancer type	Count	Proportion
BRCA	300	0.3745
KIRC	146	0.1823
LUAD	141	0.1760
PRAD	136	0.1698
COAD	78	0.0974

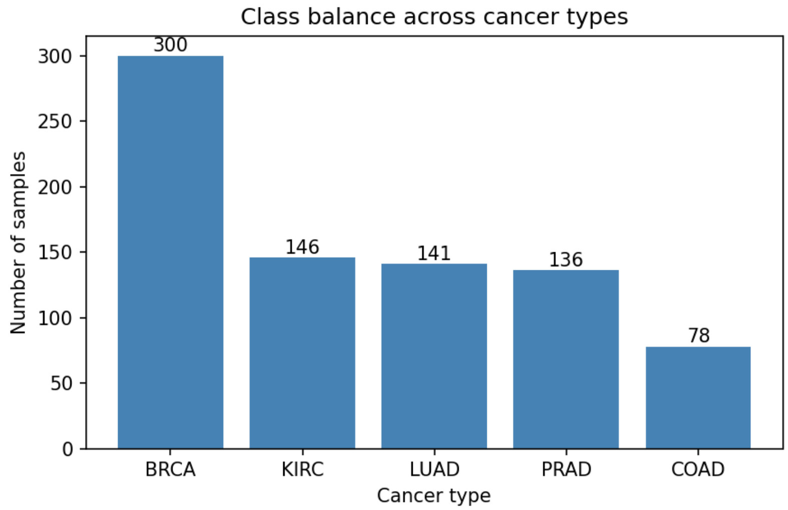

Figure 1: Sample counts per cancer type. BRCA is the majority class ($n = 300$) and COAD is the minority class ($n = 78$), a roughly 3.8× imbalance ratio.

2 Preprocessing

Preprocessing is applied in a fixed order: (1) a log transform, (2) removal of zero-variance genes, (3) standardisation, and (4) an optional variance-based top-k filter used only for computationally intensive sub-analyses.

Log transform. RNA-Seq counts are right-skewed and approximately log-normal, since gene expression is a multiplicative regulatory process rather than an additive one. Working with raw counts would let a handful of extreme-magnitude genes dominate PCA and covariance-based estimators. It would also let Euclidean-distance methods (t-SNE, UMAP, k-NN estimators, the correlation integral) have their metric set almost entirely by the highest-count genes. We therefore apply $X_{\text{log}} = \log(1 + X_{\text{raw}})$, with the +1 avoiding $\log(0) = -\infty$ while preserving relative ordering.

Zero-variance genes. After the log transform, 267 genes remain exactly constant across all 801 samples. These carry no discriminative information and are undefined under z-scoring (division by zero), so they are dropped rather than imputed with an artificial variance. No information is lost, since they are constant by definition. This leaves 20,531 − 267 = 20,264 genes, the "full" feature set used for all subsequent full-dimensional statistics.

Standardisation. Each retained gene is rescaled to zero mean, unit variance *after* the log transform rather than before. The log transform is a nonlinear step, so its effect on skew would be undone if applied to already-scaled data. Standardisation is meant to equalise per-gene scale, not to compensate for raw-count skew. This matters for two classes of downstream method. For covariance-based methods (PCA, the Marchenko–Pastur comparison), genes with the largest raw variance would dominate the sample covariance for reasons of scale alone, not biology. For distance-based methods (t-SNE, UMAP, k-NN, the correlation integral), Euclidean distance is scale-sensitive by construction.

Variance-based filtering. Some analyses, such as computing the dense empirical covariance spectrum, are intractable at the full $d = 20{,}264$. We therefore additionally save a reduced dataset containing the top $k = 2{,}000$ most-variable genes, ranked by variance on the log-transformed (pre-standardisation) data. Ranking after standardisation would be meaningless, since every standardised gene has unit variance by construction. The choice $k = 2{,}000$ is made solely for computational tractability and is not intended to imply any particular intrinsic dimensionality. Whenever computationally feasible, analyses are still performed and reported on the complete 20,264-gene set.

Figure 2 shows the per-gene mean and variance distributions (log scale, full feature set). Means are bimodal: there is a cluster of lowly or rarely expressed genes near zero, and a broad mode of moderately-to-highly expressed genes around 2–2.5. Variance is heavily right-skewed. The long tail of high-variance genes holds the best candidates for cancer-type-discriminative signal, and is exactly what the top-k filter selects (Figure 3).

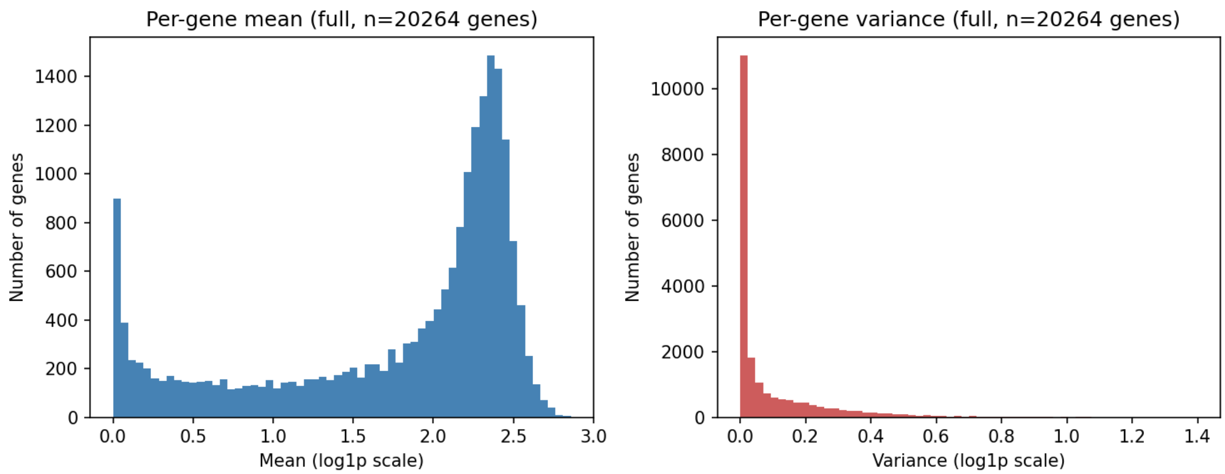

Figure 2: Per-gene mean (left) and per-gene variance (right) on the log-transformed scale, full retained feature set ($d = 20{,}264$ genes).

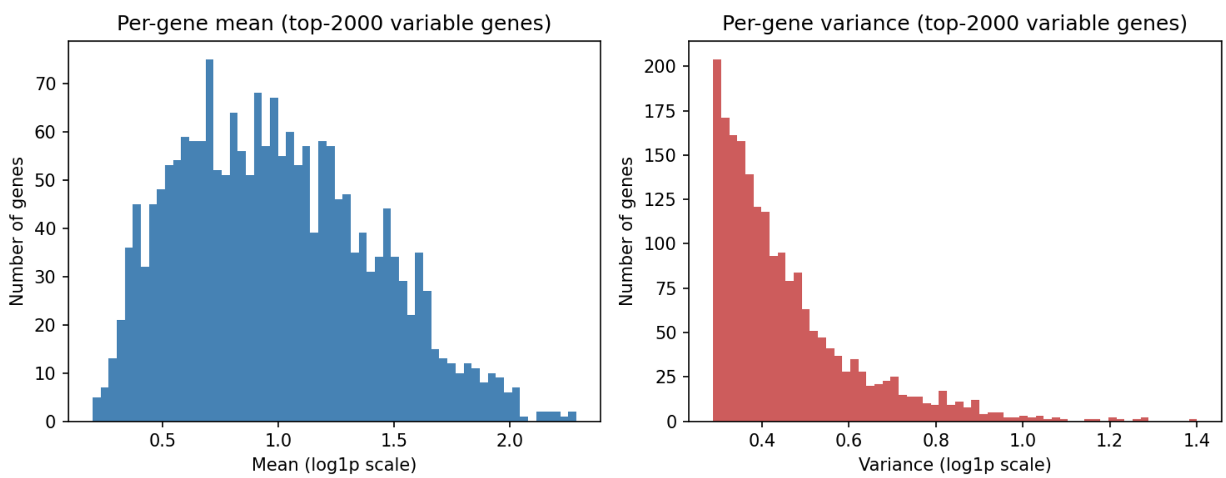

Figure 3: Per-gene mean (left) and per-gene variance (right) on the log-transformed scale, restricted to the top-2,000 most variable genes.

3 Intrinsic Dimension Estimation

All estimators here operate on the standardised $801 \times 20{,}264$ matrix. Under the small-n, large-d regime, the *ambient* dimension is a poor proxy for the *intrinsic* dimension of the data-generating manifold. Gene co-expression is highly correlated across biological pathways, so the 801 tumor samples likely occupy a much lower-dimensional structure than 20,264 free coordinates. We measure this with three estimators that rest on different mathematical assumptions (neighborhood scaling, global linear variance, and local maximum-likelihood neighbor statistics), and calibrate each against synthetic data of known ground-truth dimensionality.

3.1 Estimators

Correlation dimension. The Grassberger–Procaccia correlation integral (Grassberger and Procaccia, 1983)

$$C(r) = \frac{2}{N(N-1)} \sum_{i<j} \mathbb{1}\left[\|x_i - x_j\| < r\right] \tag{1}$$

is computed exactly from the condensed pairwise-distance vector, with $\mathbb{1}[\cdot]$ the indicator function in (1) evaluated over all $\binom{N}{2}$ pairs. D_2 is the slope of $\log C(r)$ vs. $\log r$ over an automatically detected scaling region. Candidate radii are 60 log-spaced points between the 1^{st} and 90^{th} percentile of pairwise distances. Those two quantiles are estimated from a random 300-sample subset, purely to keep radius selection cheap – the correlation integral itself is always evaluated on all 801 points. The scaling region is the window (minimum width 5 points) with the smallest local-slope variance, i.e. the most linear stretch of the log-log curve. Bounding $r_{\min}$ above the 1^{st} percentile avoids finite-sample quantisation noise. Bounding $r_{\max}$ below the 90^{th} percentile avoids saturation from the finite diameter of the ambient cloud. This selection rule searches for the most linear window available, so the resulting R^2 is close to circular: any five consecutive points on a smooth curve give a near-perfect linear fit. R^2 is reported below for completeness, but it certifies only that a locally linear window was found, not that a genuine scaling region exists.

PCA-based estimation. The covariance spectrum is obtained via the economy SVD of the centered matrix (the full $d \times d$ covariance is ~ 3.3GB and rank-deficient, and is never formed), giving eigenvalues for all 801 nontrivial components. The number of components needed for cumulative explained-variance thresholds $\{0.90, 0.95, 0.99\}$ is read directly off the spectrum. Two elbow criteria are reported. The *Kaiser criterion* ($\#\{i : \lambda_i > \bar{\lambda}\}$) is scale-free but assumes eigenvalues are comparable to their mean. The *curvature criterion* takes the index of maximum discrete second derivative of the variance-ratio curve. A scree curve is decreasing and convex, so its second difference peaks where the steep initial drop flattens out. It is sharper than Kaiser on spectra with a clean break, but it degenerates on smoothly decaying ones, where no single index stands out.

k-NN MLE. The Levina–Bickel estimator (Levina and Bickel, 2004) computes, for each sample, a local dimension estimate from neighbor-distance ratios $T_1 < \cdots < T_k$:

$$m_k(x_i) = \left[\frac{1}{k-2} \sum_{j=1}^{k-1} \ln \frac{T_k(x_i)}{T_j(x_i)}\right]^{-1}, \tag{2}$$

where $T_j(x_i)$ is the distance from x_i to its j-th nearest neighbor. Equation (2) is evaluated per sample and then averaged over all samples, swept at $k \in \{5, 10, 20\}$ to expose the bias-variance trade-off. Small k is local and low-bias but high-variance, while large k smooths over a larger neighborhood and risks bias if the neighborhood exceeds the locally linear region of a curved manifold. Local estimates that are degenerate (from tied neighbor distances, which would send the denominator to zero) are filtered out with a finiteness-and-positivity check before averaging.

3.2 Synthetic Calibration

Two synthetic datasets, matched in shape to the real matrix, are passed through all three estimators identically to the real data. The **Gaussian noise baseline** draws every coordinate i.i.d. from $\mathcal{N}(0, 1)$ (no structure, an upper-bound reference). The **linear manifold baseline** embeds 801 latent points $z_i \sim \mathcal{N}(0, I_5)$ into $\mathbb{R}^{20,264}$ via a fixed random orthonormal basis, then adds isotropic noise $\mathcal{N}(0, \sigma^2 I)$. This gives a dataset with a known ground-truth intrinsic dimension of 5.

Both baselines are then z-scored per coordinate before estimation, exactly like the real data, so all three datasets reach the estimators through the same preprocessing path. This matters for the calibration argument. Had the baselines been left unscaled while the real data was standardised, any agreement or disagreement between them would be partly a byproduct of the preprocessing rather than of the geometry.

Because the orthonormal embedding preserves the latent covariance's eigenvalues exactly (5 near 1, the rest exactly 0), adding isotropic noise of variance σ^2 raises every eigenvalue by σ^2. This contributes $d\sigma^2$ of total variance against only ≈ 5 from signal. Since $d = 20{,}264$ is enormous, this noise term dominates for any σ that looks "small" in isolation. At $\sigma = 0.1$, noise variance (≈ 203) swamps signal (≈ 5), and the estimators move sharply toward the pure-noise regime (Table 2). The Kaiser elbow reports 357 and the k-NN MLE reports 532.72. At $\sigma = 0.01$ the Kaiser elbow already recovers the true dimension exactly, while the PCA variance threshold (503) and the k-NN MLE (31.75) remain badly inflated. At $\sigma = 0.001$, all three recover something close to 5 while the perturbation remains meaningful. This is the value used for the linear-manifold baseline throughout (Table 3). The lesson is that "small noise" must be judged relative to d, not to the signal's own scale.

The correlation dimension deserves its own comment, because it does not move monotonically across the sweep: 18.06, then 1.22, then 4.29. The middle value is not a mild bias but a collapse to roughly a quarter of the truth, and the ordering does not track the noise level. This is a property of the scaling-region search rather than of the manifold. When no real power-law regime exists, the criterion still returns the flattest available window, and which window that is can change abruptly under small perturbations of the point cloud. The same fragility is visible in the Gaussian-noise baseline, whose D_2 moves from 49.02 to 419.52 under nothing more than per-coordinate standardisation, a transformation that alters pairwise distances by well under a percent. D_2 values in this report should therefore be read as order-of-magnitude indicators, and the R^2 that comes with them as a statement about the selected window, not about how stable the estimate is.

Table 2: Manifold-baseline sensitivity to additive noise level σ (true dimension = 5). PCA@90 = components for 90% cumulative variance, kNN@10 = Levina–Bickel mean at $k = 10$. The $\sigma = 0.001$ row is the configuration adopted for the linear-manifold baseline in Tables 3 and 4.

σ	D_2	Kaiser elbow	PCA@90	kNN-MLE ($k = 10$)
0.100	18.06	357	686	532.72
0.010	1.22	5	503	31.75
0.001	4.29	5	5	5.42

3.3 Results

Table 3: Intrinsic dimension estimates: real data vs. synthetic calibration baselines.

Dataset	D_2	R^2	PCA@90	PCA@95	PCA@99	kNN_5	kNN_{10}	kNN_{20}
Real (RNA-Seq)	7.73	1.000	392	548	732	33.16	29.55	25.37
Gaussian Noise	419.52	1.000	688	742	788	626.09	621.67	583.31
Linear Manifold	4.29	1.000	5	5	5	5.69	5.42	5.20

Table 4: PCA scree-plot elbow criteria per dataset (component index).

Dataset	Kaiser elbow	Curvature elbow
Real (RNA-Seq)	100	3
Gaussian Noise	386	54
Linear Manifold	5	5

Figure 4 shows explained variance decaying sharply within the first ~ 10 components, then following a long, shallow tail out to 801. This is consistent with a compact set of dominant directions plus a heavy-tailed residual spread from biological and technical variation. Of the two elbow criteria in Table 4, the curvature criterion is the one the baselines validate most cleanly. It returns exactly 5 on the linear-manifold baseline, matching both the ground truth and the Kaiser criterion. On the real data, however, it returns 3. That is not a low-dimensionality claim. It is a symptom of the criterion's failure mode: it needs a clean break in the spectrum to find, and the real scree curve $(10.44\%, 8.56\%, 7.64\%, \dots)$ simply decays smoothly. So the maximum second difference falls near the front of the curve without marking anything real. The Kaiser elbow (100) is the more defensible global-linear summary on this data, though still well above D_2 and the k-NN estimates.

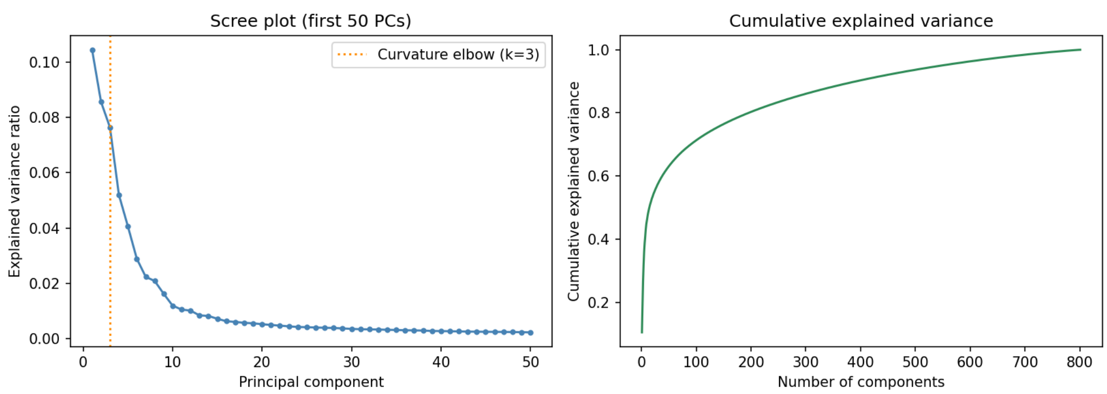

Figure 4: Scree plot of the real RNA-Seq covariance spectrum: per-component explained variance (first 50 PCs, left) and cumulative explained variance (all 801 components, right).

Figure 5 shows the log-log correlation integral. The curve is smoothly convex with no long, sharply linear stretch. The automatically detected scaling region (5 points, $r \in [149.6, 156.5]$) is narrow, which reflects both the modest sample size and the smoothly varying local density typical of standardised, correlated gene-expression data.

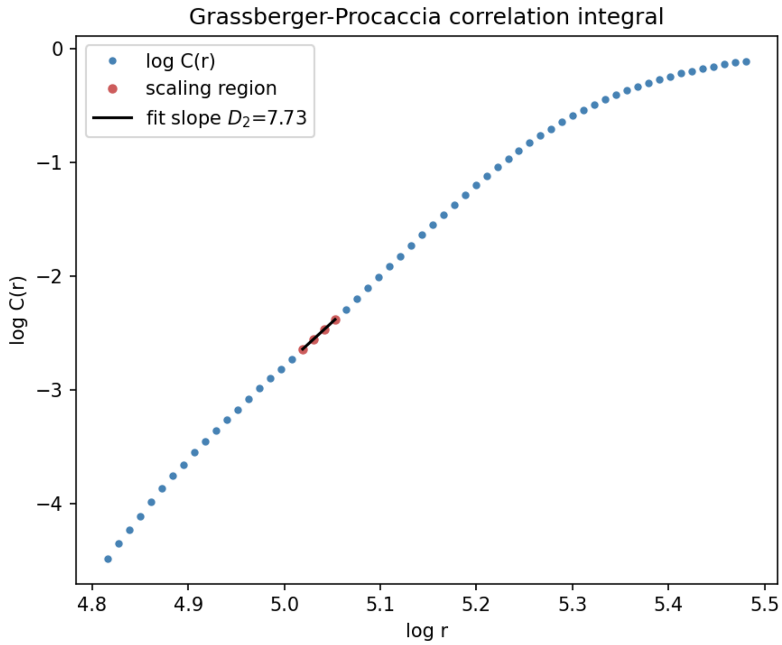

Figure 5: Log-log plot of the correlation integral $C(r)$ for the real data, with the detected scaling region and fitted slope $D_2 = 7.73$ highlighted.

Reconciling the three estimators. They disagree substantially on the real data ($D_2 \approx 7.7$, k-NN MLE mean ≈ 29.4, Kaiser elbow 100, 392 components for 90% variance). The synthetic baselines show this disagreement is expected rather than an implementation error, since all three converge tightly on the correct answer (≈ 5) for the linear manifold once noise is controlled. D_2 and k-NN MLE are *local* estimators, inferring dimension from how neighbor statistics scale, which is sensitive to distance sparsity when n is small relative to d. That is consistent with their well-documented negative bias in this regime. It also explains why D_2 falls below the kNN-MLE value here: D_2's scaling region sits at a coarser radius scale, with fewer effective neighbor pairs. PCA, in contrast, is a *global, linear* estimator. It counts every direction of non-negligible variance. This mixes together true low-dimensional nonlinear structure, which a linear method must represent with many components, and real additional biological variance, such as subtype heterogeneity within each cancer type, plus technical variance – both of which the local estimators are comparatively insensitive to. The gap between the Kaiser elbow (100) and the 90%-variance threshold (392) shows a further effect: raw variance thresholds are dominated by the combined mass of the long eigenvalue tail, even when individual tail eigenvalues are small. This is the same mechanism the noise sweep (Table 2) demonstrates directly.

Together, the real data's estimates place the intrinsic dimension of the Pan-Cancer expression manifold far below $d = 20{,}264$: two to three orders of magnitude below it by the local estimators (single digits to a few dozen), and still roughly two orders below it by the global Kaiser criterion (low hundreds). All of these values sit far closer to the number of cancer types (five) than to d. This fits with gene co-expression collapsing the effective degrees of freedom of a tumor's transcriptomic profile well below the number of measured genes.

4 Geometry, Visualisation, and Random-Matrix Diagnostics

4.1 PCA, t-SNE, and UMAP

The first 3 PCs, computed via the exact (`svd_solver="full"`) SVD for byte-for-byte reproducibility, are used for 2D/3D scatter plots (Figures 6 and 7).

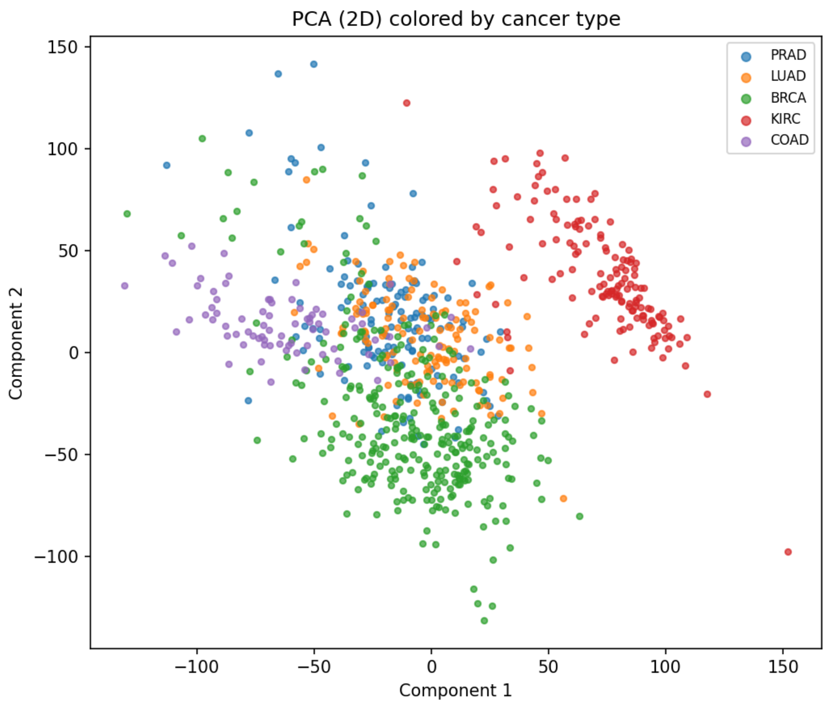

Figure 6: PCA 2D projection, colored by cancer type. PC1 explains 10.44% of variance, PC2 8.56%, cumulative 19.00%.

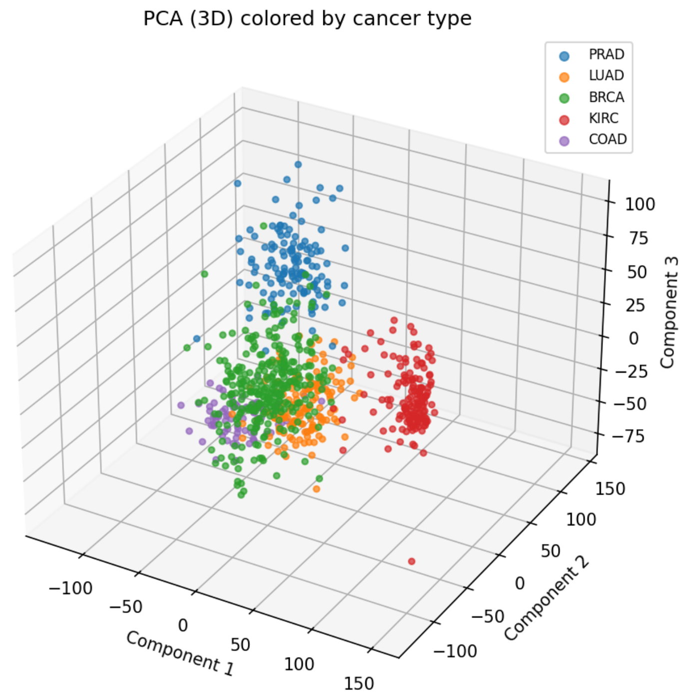

Figure 7: PCA 3D projection. PC3 adds 7.64%, for a cumulative 26.64% across all three.

Even with only 26.64% of total variance captured, KIRC and COAD form visually distinct clusters along PC1/PC2, while BRCA, LUAD, and PRAD show only partial linear separability with substantial overlap. This is consistent with PCA being a purely linear, global estimator that spreads nonlinear cluster boundaries across many components.

t-SNE (van der Maaten and Hinton, 2008) at perplexity 30 and 50, and UMAP (McInnes et al., 2018) at `n_neighbors` = 15 and 50, were both preprocessed with a 50-D linear PCA reduction. That step denoises minor components and reduces neighbor-search cost from $O(n^2 d)$ to $O(n^2 \times 50)$. Fifty components comfortably exceed the range the local estimators in Section 3 attribute to real structure (single digits to a few dozen). Note that 50 was fixed in advance as a standard default rather than tuned. No sweep over the preprocessing dimension was run, so we can say that 50 components are enough, but not that fewer would fail (Section 6).

Both methods produce five visually well-separated groups at every hyperparameter tested (Figures 8 and 9). This is a sharp contrast with the muted linear separation above, and it suggests the cluster structure is real but strongly nonlinear. Changing perplexity or `n_neighbors` mainly rescales and repositions the groups rather than reorganising them. This assessment is visual. No clustering metric was computed, so the correspondence between the visible groups and the true cancer-type labels is not measured here, and claims below that rest on it should be read with that caveat. Computing an adjusted Rand index against the labels, and between hyperparameter settings, would settle both points and is the first item of future work (Section 6).

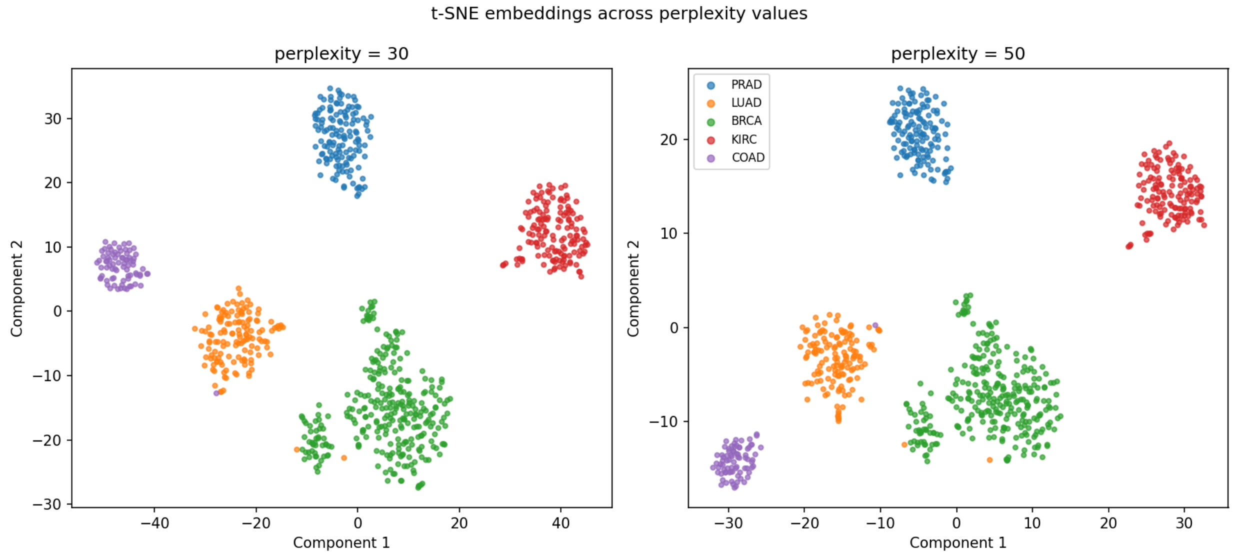

Figure 8: t-SNE 2D embeddings at perplexity 30 (left) and 50 (right), both PCA-preprocessed to 50 dimensions, colored by cancer type.

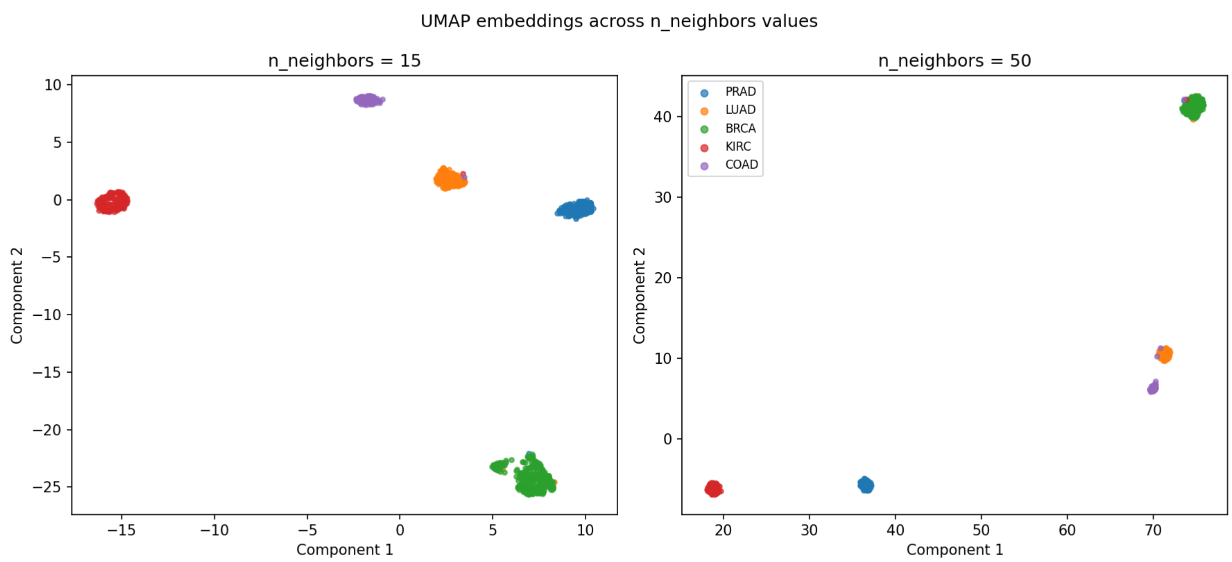

Figure 9: UMAP 2D embeddings at `n_neighbors` $= 15$ (left) and `n_neighbors` $= 50$ (right), both PCA-preprocessed to 50 dimensions, colored by cancer type.

4.2 Johnson–Lindenstrauss Random Projection

The JL lemma (Johnson and Lindenstrauss, 1984) guarantees a linear map into $\mathbb{R}^k$ preserving all pairwise distances up to $(1 \pm \varepsilon)$ for

$$k \geq \frac{4 \ln n}{\varepsilon^2/2 - \varepsilon^3/3} \tag{3}$$

(Dasgupta and Gupta, 2003), achieved with high probability by a random Gaussian matrix $R_{ij} \sim \mathcal{N}(0, 1/k)$. Evaluating (3) at $n = 801$, the conservative default $\varepsilon = 0.1$ gives $k \geq 5{,}731$, which barely compresses $d = 20{,}264$. We instead use $\varepsilon = 0.2$, giving $k = 1{,}543$, a $13.1\times$ reduction at what is still a standard distortion tolerance.

Table 5: Johnson–Lindenstrauss random projection: target dimension and empirical distance-preservation summary at $\varepsilon = 0.2$.

Quantity	Value
Ambient dimension d	20,264
JL target dimension k	1,543
Compression ratio d/k	$13.13\times$
Mean distortion ratio $\bar{\rho}$	0.9996
Max $\lvert\rho - 1\rvert$ (empirical)	0.0808
Fraction of pairs within $[1-\varepsilon, 1+\varepsilon]$	1.0000

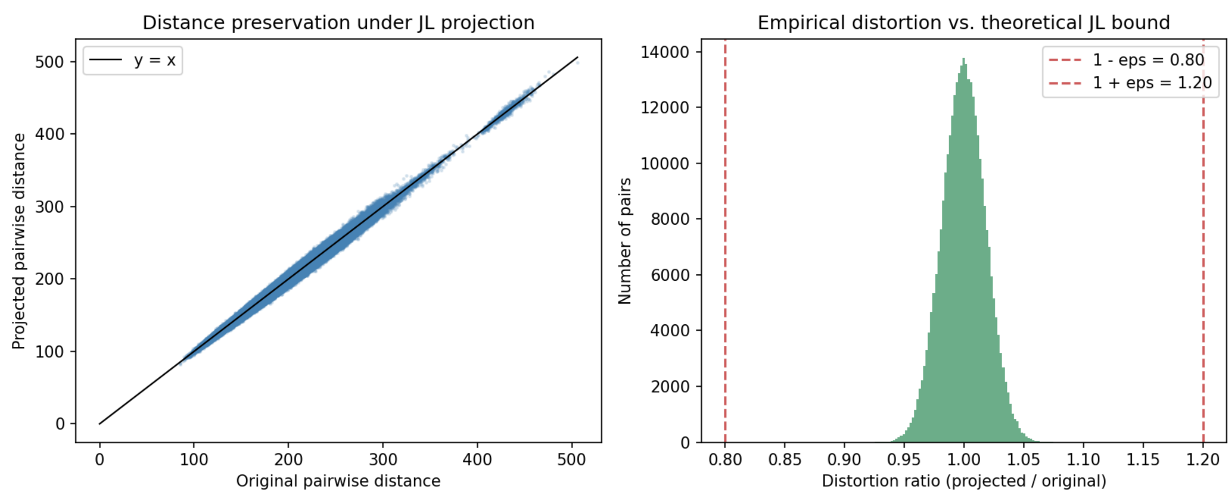

Figure 10: Left: projected vs. original pairwise distances for all $\binom{801}{2} = 320{,}400$ pairs, with the identity line $y = x$. Right: histogram of the distortion ratio ρ_{ij}, with the theoretical $(1 \pm \varepsilon)$ bound marked.

Every one of the 320,400 pairwise distances satisfies the theoretical guarantee (Table 5, Figure 10). The empirical distortion is tightly concentrated around 1.0 with a maximum deviation of 0.081, roughly $2.5\times$ tighter than the bound requires, so the projection is conservative in practice.

It is worth being precise about what this does and does not show. The JL guarantee is a statement about the random projection, not about the data: it holds for *any* set of 801 points in *any* ambient dimension, so isotropic noise would satisfy it exactly as well as this dataset does. This experiment therefore validates the implementation and demonstrates the lemma at a realistic scale. It also establishes that a $13.1\times$ compression is available at negligible metric cost for any downstream distance-based method. It is not evidence that the data are low-dimensional, and it is not counted as such in Section 5.

4.3 Marchenko–Pastur Spectral Analysis

For a $p \times n$ matrix of i.i.d. unit-variance entries with $p/n \to \gamma$, the Marchenko–Pastur law (Marchenko and Pastur, 1967) gives the limiting spectral density

$$f_\gamma(x) = \frac{1}{2\pi\gamma x}\sqrt{(b-x)(x-a)}, \qquad a \le x \le b, \tag{4}$$

with $a = (1-\sqrt{\gamma})^2$ and $b = (1+\sqrt{\gamma})^2$. For $\gamma > 1$ (rank-deficient, as here), (4) is supplemented by a point mass $1 - 1/\gamma$ at $x = 0$. We compute the empirical spectrum of the top-2,000-gene correlation matrix (a $\sim$ 32MB matrix, trivial to diagonalise exactly, versus $\sim$ 3.3GB for the full gene set). The aspect ratio is $\gamma = 2{,}000/800 = 2.500$. The denominator is $n-1$, not n, because standardisation centers each column, which costs one degree of freedom. Using n instead would put γ at 2.497 and misstate the point-mass prediction by enough to obscure what is in fact an exact agreement. Eigenvalues exceeding b cannot be explained by an unstructured random matrix and are interpreted as spectral outliers – evidence of genuine covariance structure.

One caveat applies to that interpretation throughout. The 2,000 genes are not a random subset. They are the highest-variance genes on the log scale, and high variance in expression data comes largely from coordinated regulation across co-expression modules, which is exactly the structure that produces large eigenvalues. The MP null, by contrast, assumes an unselected matrix. The outlier count below is therefore an upper-biased estimate *conditional on that selection*, and is not the number the full 20,264-gene matrix would give. It is evidence that structure exists, and that is what it is used for here. It is not a measurement of how much.

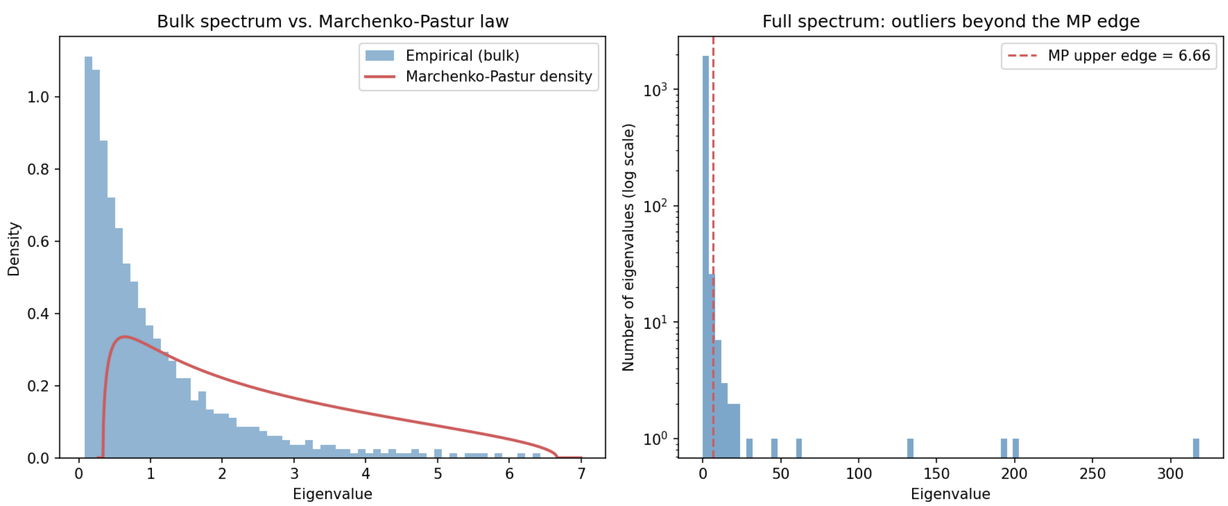

Figure 11: Left: empirical bulk spectrum vs. the Marchenko–Pastur density ($\gamma = 2.500$), rescaled to the conditional (nonzero) distribution. Right: full spectrum on a log-count axis, showing 27 outlier eigenvalues far beyond the theoretical upper edge (dashed line).

Table 6: Ten largest spectral outlier eigenvalues (top-2,000-gene correlation matrix, $\gamma = 2.500$, theoretical support $[0.338, 6.662]$). Their sum, 1,043.09, is 52% of the matrix trace (2,000).

Rank	Eigenvalue	Rank	Eigenvalue
1	318.34	6	46.16
2	202.87	7	27.95
3	191.98	8	22.69
4	131.53	9	20.10
5	62.10	10	19.37

The point-mass prediction is confirmed exactly. 60.00% of the 2,000 eigenvalues are numerically zero, against a theoretical $1-1/\gamma = 60.00\%$. Of the 800 nonzero eigenvalues, 27 (3.4% of the nonzero spectrum, 1.35% of all 2,000) exceed the bulk edge $b = 6.662$, the largest reaching 318.3, some 48× the edge.

The bulk histogram (Figure 11, left) overlaps the Marchenko–Pastur support broadly, but it is visibly more concentrated at small eigenvalues than theory predicts. The reason is arithmetic, not mysterious. A correlation matrix has trace exactly $p = 2{,}000$, so the spectrum is a fixed budget that the outliers and the bulk must share. The ten largest outliers alone (Table 6) sum to 1,043.09, which is 52% of the entire trace. That leaves at most 957 to be divided among the remaining 790 nonzero eigenvalues, an average of at most ≈ 1.2, compared with the Marchenko–Pastur conditional mean of $\gamma = 2.5$. A bulk compressed by roughly a factor of two toward zero is therefore exactly what the outliers force, and the discrepancy is not independent evidence of anything beyond the outliers themselves. A sharper comparison would refit the bulk against an effective σ^2 estimated from the post-outlier trace rather than against the unit-variance null.

4.4 Distance Concentration

We compute the relative spread ratio $(\max - \min)/\min$ of pairwise distances at feature-subsample sizes $\{10, 100, 1{,}000, 20{,}264\}$, compared against the Chernoff-type bound of Laurent and Massart (2000) for $X \sim \chi^2_d$,

$$P\left(\left|\frac{X}{d} - 1\right| \geq \varepsilon\right) \leq 2\exp\left(-d\left(\frac{\sqrt{1+2\varepsilon}-1}{2}\right)^2\right), \tag{5}$$

evaluated at $\varepsilon = 0.1$. The form in (5) was chosen over a literal Hoeffding bound because standardised RNA-Seq features are effectively unbounded (they are approximately Gaussian after the log transform). The chi-squared/Laurent–Massart form is the right Chernoff-type analogue for sums of squared sub-Gaussian coordinates.

Table 7: Pairwise distance concentration ratio and theoretical bound across feature-subsample sizes.

Feature count	$(\max - \min)/\min$	Theoretical bound $P(\lvert X/d - 1\rvert \geq \varepsilon)$
10	51.79	1.000
100	6.29	1.000
1,000	5.24	0.2051
20,264 (all)	4.96	1.81×10^{-20}

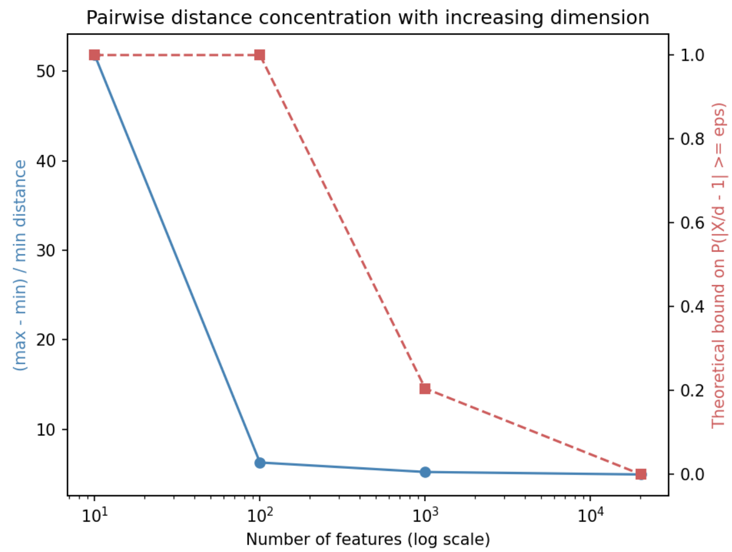

Figure 12: Relative distance-spread ratio (blue, left axis) and theoretical tail bound (red, right axis) vs. number of features, log-scaled x-axis.

The ratio (Table 7, Figure 12) drops sharply from 51.79 at 10 features to 6.29 at 100, then flattens to 5.24 at 1,000 features and 4.96 at all 20,264. Each row is a single random draw of that many genes. At $d = 10$, one draw out of 20,264 carries a lot of sampling variability, so the first row should be read as a rough indication of size, not a precise value. The theoretical bound is essentially meaningless at $d = 10, 100$, reaches 0.205 at $d = 1{,}000$, and drops to 1.81×10^{-20} at $d = 20{,}264$.

The two columns are not directly comparable, and the table should not be read as a like-for-like test. The bound is a tail probability for a *single* pair's squared distance under an i.i.d. Gaussian model, whereas the ratio is a max-over-minimum statistic across all 320,400 pairs of unsquared distances. Making the comparison rigorous requires a union bound over all pairs, 3.204×10^5 times the per-pair bound. That bound is essentially meaningless at every feature count except the full set, where it still gives 5.8×10^{-15}. So at full dimension the i.i.d. model does predict that essentially no pair deviates by more than 10% in squared distance, while the observed spread ratio is 4.96. Read that way, the qualitative conclusion survives: gene co-expression, rather than the i.i.d. behavior the bound assumes, is the natural explanation for why concentration stalls. But the mismatch is a comparison of a bound against a different statistic, and it is weaker evidence than a matched test would be.

4.5 Ledoit–Wolf Shrinkage and Mahalanobis Anomaly Detection

Ledoit–Wolf shrinkage estimates covariance as $\hat{\Sigma}_{\mathrm{LW}} = (1-\alpha)S + \alpha\mu I$, $\mu = \mathrm{tr}(S)/p$, with α chosen analytically (Ledoit and Wolf, 2004) to minimise expected Frobenius loss. Unlike S itself, which is exactly singular whenever $p \geq n$, $\hat{\Sigma}_{\mathrm{LW}}$ is guaranteed invertible. The squared Mahalanobis distance $D^2(x_i) = (x_i - \bar{x})^\top \hat{\Sigma}_{\mathrm{LW}}^{-1}(x_i - \bar{x})$ is computed on the top-2,000-gene subset and flagged as an outlier if it exceeds the 97.5% quantile of a χ^2 reference.

The degrees of freedom of that reference require care, and getting them wrong makes the test meaningless. The natural choice is the feature count $p = 2{,}000$, one per coordinate. But with $p > n$ the centered data occupy a subspace of dimension only $n - 1 = 800$, and D^2 is confined to it. Writing $\ell_1, \ldots, \ell_{800}$ for the nonzero eigenvalues of the sample covariance S and noting that $\mu = \mathrm{tr}(S)/p = 1$ because the columns are standardised,

$$\frac{1}{n}\sum_{i=1}^{n} D^2(x_i) = \sum_{j=1}^{n-1} \frac{\ell_j}{(1-\alpha)\ell_j + \alpha\mu} \;\leq\; \frac{n-1}{1-\alpha}, \tag{6}$$

so the mean squared distance is bounded above by 819.6 at the fitted $\alpha = 0.0239$, and tends to exactly 800 as $\alpha \to 0$, no matter what the data are. The 97.5% quantile of χ^2_{2000} is 2,125.84, well over twice that ceiling, so a p-df threshold cannot be crossed by any dataset of this shape. We therefore use $\mathrm{df} = n-1 = 800$, the rank of the estimator, giving a threshold of 880.28. This remains an approximation. Because $\bar{x}$ and $\hat{\Sigma}_{\mathrm{LW}}$ are estimated from the same 801 samples being scored, the exact reference for D^2 is a scaled Beta distribution rather than a χ^2, and shrinkage perturbs it further. It is, however, an approximation a sample can actually fail.

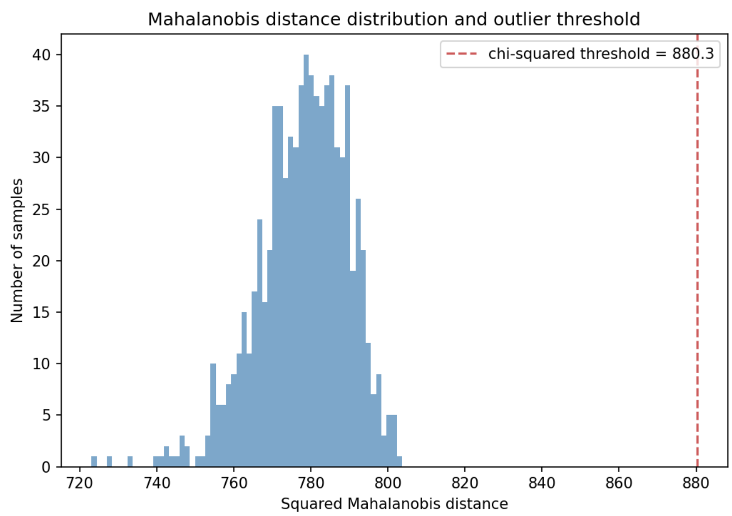

Figure 13: Distribution of squared Mahalanobis distances (top-2,000-gene subset, Ledoit–Wolf shrinkage) across all 801 samples, with the χ^2_{800} 97.5%-quantile outlier threshold marked.

Table 8: Ten samples with the largest squared Mahalanobis distance (none exceed the outlier threshold of 880.28). The distances are separated by hundredths, so the ordering within this list is not meaningfully resolved.

Sample	Cancer type	Squared Mahalanobis distance
sample_560	BRCA	803.66
sample_250	BRCA	801.93
sample_599	LUAD	801.87
sample_322	KIRC	801.85
sample_789	KIRC	801.69
sample_497	BRCA	801.20
sample_577	LUAD	800.19
sample_389	BRCA	799.99
sample_77	KIRC	799.92
sample_343	LUAD	799.92

The shrinkage intensity is $\hat{\alpha} = 0.0239$, which is light but sufficient to guarantee invertibility. Squared Mahalanobis distances (Figure 13) range from 723.10 to 803.66 (mean 778.10, std 11.57), below the χ^2_{800} threshold of 880.28, so **zero of 801 samples are flagged as outliers**.

Two things are worth separating here. The first is that the observed distances sit exactly where equation (6) says they must: a mean of 778.10 against a ceiling of 819.6, with a standard deviation of only 11.57, i.e. a relative spread of 1.5%. That tightness is structural, not a property of these tumors. It is the $p > n$ geometry showing through: in the metric induced by a rank-deficient covariance estimate, every sample is almost exactly as far from the centroid as every other. Under the original χ^2_{2000} threshold this produced a guaranteed null result. Under the corrected χ^2_{800} threshold the most extreme sample reaches 91.3% of the cutoff, so the test now has room to fire and does not.

The second is what that null result actually tells us. It is not "the data has no outliers". It says that no sample is extreme relative to the shrunken covariance. Because the Ledoit–Wolf target is isotropic while the true spectrum is extremely heavy-tailed (Table 6, ranging from near-zero to 318.3), the blending inflates estimated variance along the many near-zero directions more than it deflates it along the few dominant ones. Deviations that lie along truly informative directions are therefore down-weighted relative to an unregularised metric. Combined with the structural compression above, this makes Mahalanobis distance here a better *relative ranking* tool than a formally calibrated test. The ten most extreme samples (Table 8) are 4 BRCA, 3 LUAD, and 3 KIRC, with no PRAD or COAD. Little should be read into that. The ten distances span 3.7 units out of ≈ 800, so which samples make this list is decided at the level of numerical noise, and with only 10 samples no class comparison is statistically robust. It is noted descriptively only.

5 Discussion: Reconciling Ambient, Intrinsic, and Visual Dimensionality

Three calibrated estimates of intrinsic dimension are available: $D_2 = 7.73$, a k-NN MLE mean of 29.36, and the PCA criteria (Kaiser elbow 100, with 392 components for 90% variance). These are not three attempts at the same measurement that failed to agree. They are three *different* operational definitions of "dimension," and their ordering is informative. D_2 is the most locally defined estimator, and the most negatively biased in this small-n, large-d regime. Its scaling region spans only 5 radii, because pairwise distances in 20,264 dimensions are sparse and concentrated (concentration ratio 4.96 at full dimension). This starves the small-radius regime of the many-pair statistics the estimator needs, and makes D_2 the most conservative of the three. The Kaiser elbow (100) and the 90%-variance threshold (392) are both computed on the full, unfiltered gene matrix. There, thousands of low-signal genes each contribute a small but nonzero amount of variance (subtype heterogeneity, technical noise), and a linear, global method cannot tell this apart from real manifold curvature. The gap between the two, a factor of nearly 4 on the same spectrum, shows how much of that count is aggregate tail mass rather than dominant directions.

The Marchenko–Pastur spectral outlier count sits outside this comparison, and it is worth saying explicitly why, because its numerical proximity to the k-NN MLE mean invites a stronger reading than it can support. It is not an intrinsic-dimension estimator. It counts directions of *linear* correlation that a null random matrix cannot produce, it was never run against the two synthetic baselines, and its value depends directly on the gene-subset size k, since both the aspect ratio $\gamma = k/(n-1)$ and the bulk edge $b = (1 + \sqrt{\gamma})^2$ are functions of k. Section 2 adopts $k = 2{,}000$ for computational tractability and is explicit that this is not a statement about dimensionality. That same caveat applies here. The outlier count is reported below as independent evidence that the correlation structure is far from random, which is a real and separate finding, and not as a fourth estimate of the same quantity. Establishing whether 27 is stable in k would require the sweep listed in Section 6.

There is a naive expectation that a small intrinsic dimension implies a similarly small number of PCs should be enough to visually separate the five cancer types. The results in Section 4 directly falsify this, and that falsification is itself central to reconciling the estimators. The first 3 PCs together explain only 26.64% of variance, and while KIRC and COAD separate visually, BRCA, LUAD, and PRAD show substantial overlap. Yet t-SNE and UMAP, applied to the same data after only a 50-D linear PCA preprocessing step, separate the five types visually at every hyperparameter tested. The resolution is that "intrinsic dimension," as estimated by D_2 and the k-NN MLE, measures the *local* degrees of freedom of the manifold: how many coordinates parameterise a small neighborhood of any given sample. It says nothing about whether that manifold is *linearly flat*. A manifold can have intrinsic dimension 7 while being curved enough through the ambient space that no 7- or even 100-dimensional linear subspace captures it via orthogonal projection with clean class separation. PCA must spend components *unfolding* curvature, not only tracking true degrees of freedom. The fact that 50 linear components give t-SNE and UMAP enough to work with to separate the classes puts a useful upper bound on where the structure lives. It is the same order of magnitude as the Kaiser elbow (100) and well below the 90%-variance threshold (392). So the curved manifold's structure sits within a linear subspace of at most ~ 100 dimensions, even though its curvature-aware intrinsic dimension is an order of magnitude smaller still ($\approx$ 7–30). It is an upper bound only. Fifty was a fixed default and was never reduced to find the point of failure, so nothing here shows that 50 is the minimum, and the bridge between the local and global estimates is correspondingly loose.

Table 9 brings together every quantitative diagnostic computed in this project. Rows populated for all three datasets show the same qualitative pattern. The Gaussian-noise baseline sets a ceiling: correlation dimension in the hundreds (419.52), k-NN MLE likewise (610.36), and hundreds of PCA components for any variance threshold, since isotropic noise has no compressible structure. None of these reaches $d = 20{,}264$, and none should be expected to. With only 801 points, every estimator here saturates well below the ambient dimension. What matters is that all three sit two orders of magnitude above the corresponding real-data values. The linear-manifold baseline recovers its known ground truth of 5 almost exactly once noise is calibrated. The real data sits strictly between the two, but far closer to the manifold regime, which is calibrated evidence that the Pan-Cancer expression manifold is genuinely low-dimensional rather than high-dimensional noise with superficial class labels. The rows populated only for the real dataset (JL, spectral, Mahalanobis) are three different kinds of evidence, and only one of them relates to dimensionality. The spectral row does. The correlation structure contains 27 directions that no random matrix of matching shape could produce, which is independent evidence that the covariance is truly structured and not noise. The JL row does not, since that guarantee holds for any point set whatsoever. It demonstrates that a 13.1× compression preserves this dataset's metric almost perfectly ($\bar{\rho} = 0.9996$), which is useful to know but is just as true of noise. The Mahalanobis row does not either. As Section 4 explains, the test as configured could not have flagged a sample under any data whatsoever, so its zero result is a property of the estimator's geometry, not a finding about tumors.

Table 9: Consolidated high-dimensional geometric diagnostics across the real TCGA Pan-Cancer dataset and two synthetic calibration baselines. "–" = not computed for that baseline. The three estimators in the upper block were run on all three datasets along one identical preprocessing path. The JL, spectral, and Mahalanobis diagnostics in the lower block were run only on the real data; they are not intrinsic-dimension estimates and are not calibrated against the baselines, as Section 5 sets out.

Metric / Estimator	**TCGA PANCAN**	**Gaussian Noise**	**Linear Manifold**
Ambient dimension d (raw, pre-filter)	20,531	–	–
Analysis dimension d (post zero-var. filter)	20,264	20,264	20,264
True intrinsic dimension (by construction)	unknown	$\approx d$ (none)	5
Correlation dimension $\hat{D}_2$ (R^2)	7.73 (1.000)	419.52 (1.000)	4.29 (1.000)
k-NN MLE mean, $k \in \{5, 10, 20\}$	29.36	610.36	5.44
Kaiser-criterion PCA elbow (components)	100	386	5
Curvature-criterion PCA elbow (components)	3	54	5
PCA components for 90/95/99% variance	392/548/732	688/742/788	5/5/5
MP spectral outlier count (top-2,000 genes)	27	–	–
JL target dimension k ($\varepsilon = 0.2$)	1,543	–	–
JL empirical mean distortion $\bar{\rho}$	0.9996	–	–
JL max $\lvert\rho - 1\rvert$ (theoretical bound: 0.2)	0.0808	–	–
Mahalanobis-flagged outliers ($\alpha = 0.025$, df = 800)	0	–	–

Conclusion. Across three independent estimators and two calibrated synthetic baselines, the TCGA Pan-Cancer expression manifold's true, curvature-aware degrees of freedom fall in the range of roughly 7 to 30 – two to three orders of magnitude below the analysis dimension $d = 20{,}264$ (raw ambient 20,531). A naive linear-PCA measure of "dimension needed for visual separation" reports a much larger number (100–392 components), because linear methods must spend components unfolding curvature that local, nonlinear-aware estimators see through directly. The reconciliation itself, rather than any single number in isolation, is the project's central technical finding. Two results support this from outside the estimator family. The nonlinear embeddings separate all five types from a 50-dimensional linear substructure that PCA alone cannot separate in three components, and the Marchenko–Pastur analysis identifies 27 directions of correlation that no random matrix of matching shape could produce. The confidence this warrants is bounded by what was not done. The embedding separation is assessed visually rather than by a clustering metric. The 50-dimensional figure is an upper bound that was never reduced to its minimum. And the count 27 is conditional on a variance-selected gene subset whose size was chosen for tractability. Each is a specific, cheap experiment, and each is listed in Section 6.

6 Limitations and Future Work

Estimator-specific limitations. D_2 is the least stable quantity in this report, and the instability is demonstrated rather than merely suspected. Its scaling region on the real data is narrow (5 points, the minimum the search allows). Because that search returns the most linear window available whether or not a real power-law regime exists, the accompanying $R^2 = 1.000$ certifies the window, not the estimate. Section 3 shows two direct consequences: D_2 is non-monotone in the manifold baseline's noise level (18.06, 1.22, 4.29), and on the Gaussian-noise baseline it moves from 49.02 to 419.52 under per-coordinate standardisation alone. A bootstrap over sample subsets, testing whether the scaling-region *location* is stable and not only the slope fitted within it, is the minimum needed before $D_2 = 7.73$ could be quoted to two decimal places as this report does. The Kaiser and curvature elbow criteria are heuristics without a formal error rate. The curvature criterion recovers the ground truth exactly on the manifold baseline but returns 3 on the real data, where the smoothly decaying spectrum gives it no break to find. The linear-manifold baseline validates recovery of a *linear* 5-D ground truth, but it does not test recovery of a curved (nonlinear) low-dimensional manifold, which is the more realistic model for biological data.

Claims resting on unmeasured quantities. Three assertions in this report are supported by inspection rather than by a computed statistic, and are flagged here so they are not mistaken for measured results. First, the five-group separation in the t-SNE and UMAP embeddings is a visual judgement: no clustering metric was computed, so neither the agreement between the visible groups and the cancer-type labels nor the stability of group membership across hyperparameter settings is quantified. Second, the 50-dimensional PCA preprocessing that precedes those embeddings was fixed in advance rather than tuned, so 50 is an upper bound on where the separable structure lives and not a minimum. Third, the Marchenko–Pastur outlier count of 27 is conditional on the top-2,000 variance-selected genes. Since the aspect ratio and bulk edge both depend on that choice, and since variance selection preferentially keeps co-expressed genes, the count is upward-biased relative to an unselected subset of the same size, and is not the full-genome value.

The theoretical bounds used in Section 4 all assume some form of independence that real gene-expression data violates by construction. The Marchenko–Pastur law and the Laurent–Massart concentration bound are both derived for i.i.d. (or uncorrelated) coordinates, whereas genes are organised into co-expressed pathways with strong pairwise and higher-order correlations. That is not a flaw, since comparing real data against these null models is exactly the point. The 27 spectral outliers and the plateauing distance-concentration ratio are direct, quantified measurements of how far gene correlation pushes the data from the i.i.d. null, rather than artifacts to be corrected. The Johnson–Lindenstrauss guarantee, by contrast, makes no distributional assumption about the data itself (only about the random projection matrix). That is exactly why its empirical satisfaction, however exact, carries no information about this dataset. It would hold equally for the Gaussian-noise baseline, and Section 4 treats it as a validation of the implementation rather than as a result. The Mahalanobis χ^2 threshold assumes asymptotic multivariate normality of D^2. Beyond that, because the mean and covariance are estimated from the same samples being scored, the exact reference is a scaled Beta rather than a χ^2 at any degrees of freedom, and a permutation-based or bootstrap threshold would be more defensible with more compute budget. Finally, the 50-D PCA preprocessing before t-SNE and UMAP, while standard practice, could mask nonlinear structure concentrated in later linear components. A direct comparison against unpreprocessed embeddings was not run here for computational-budget reasons.

Small per-class sample sizes. BRCA accounts for 300 of 801 samples (37.45%) while COAD accounts for only 78 (9.74%). Every class-conditional claim made in this project, including the visual cluster-purity assessment and the class breakdown of the ten most-extreme Mahalanobis-distance samples (4 BRCA, 3 LUAD, 3 KIRC, 0 PRAD/COAD), carries much more sampling uncertainty for COAD than for BRCA. With $n_{\text{COAD}} \approx 78$, any claim that COAD samples are or are not disproportionately represented among outliers has wide confidence intervals under a binomial or hypergeometric null and should not be over-interpreted. The absence of COAD in the top-10 list is at least partially explained by its small class share alone (9.74%, so ≈ 1 COAD sample is expected in a random top-10 by chance).

Potential technical batch effects. TCGA data are aggregated across dozens of contributing sequencing centers over several years, with instrument and sample-preparation protocols not held constant across submissions. No batch or center covariate is present in this Kaggle-mirrored release, so batch effects cannot be directly tested or corrected here. This is an important caveat for every geometric claim above, because sequencing-depth normalisation differences, reagent-lot effects, or center-specific pipelines could add systematic, non-biological variance that a purely unsupervised geometric analysis cannot tell apart from real cancer-type signal. Since cancer type is likely correlated with submission center, some fraction of the very clean t-SNE and UMAP separation may be inflated by a batch-cancer-type confound. This does not invalidate the intrinsic-dimension estimates, since batch effects are themselves a lower-dimensional, systematic source of variance and would, if anything, tend to *reduce* rather than inflate the estimated intrinsic dimension. It does mean that "intrinsic dimension of the tumor manifold" as measured here should be read as "intrinsic dimension of the tumor-plus-processing-batch manifold," a distinction that cannot be resolved without batch metadata this dataset does not provide.

Future work. The first four items below close the gaps flagged above. They are individually cheap and would turn three visual or conditional claims into measured ones. The rest need more time or external data.

(1) *Quantify the cluster separation.* Run k-means at $k = 5$ on each t-SNE and UMAP embedding and report the adjusted Rand index against the cancer-type labels, plus the ARI between the two perplexity settings and between the two `n_neighbors` settings. The first number replaces the visual separation judgement with a measurement. The second directly tests the membership-stability claim this report does not make. (2) *Find the minimum preprocessing dimension.* Sweep the PCA preprocessing dimension over $\{5, 10, 25, 50, 100\}$ and unpreprocessed, and locate where separation degrades. This would turn the 50-dimensional upper bound into a two-sided bracket and tighten the bridge between the local and global estimates in Section 5. (3) *Test the spectral count's dependence on gene selection.* Recompute the Marchenko–Pastur outlier count at $k \in \{1000, 2000, 4000\}$, and against a *random* 2,000-gene subset as a control. The former tests whether 27 is stable in k. The latter isolates how much of the count comes from variance selection rather than from biology. (4) *Bootstrap stability selection for D_2*, repeating the full correlation-dimension pipeline across, for example, 200 bootstrap resamples of the 801 patients, to obtain a confidence interval and test whether the scaling-region location itself is stable. Given the instability documented above, this is the item most likely to change a headline number. (5) Replicate the distance-concentration diagnostic over ~ 20 random feature draws per subsample size, reporting an interquartile band rather than a single draw. (6) Nonlinear (eigenvalue-dependent) shrinkage covariance estimation (Ledoit and Wolf, 2017) in place of a single scalar intensity, and a permutation-calibrated Mahalanobis threshold in place of the asymptotic χ^2 approximation, to test whether the null anomaly result survives a correctly calibrated test. (7) A curved (nonlinear) synthetic calibration baseline, for example a Swiss roll or S-curve embedded with matched noise in $\mathbb{R}^{20,264}$, to calibrate every estimator's behavior under curvature directly rather than inferring it indirectly. (8) Obtaining true batch and center metadata from the original TCGA/GDC portal and applying a standard batch-correction method (e.g., ComBat, Johnson et al., 2007) before re-running the full pipeline.

References

Gene expression cancer rna-seq (tcga pan-cancer hiseq). `https://www.kaggle.com/datasets/debatreyadas/gene-expression-cancer-rna-seq`. Dataset accessed via Kaggle.

S. Dasgupta and A. Gupta. An elementary proof of a theorem of johnson and lindenstrauss. *Random Structures & Algorithms*, 22(1):60–65, 2003.

S. Fiorini. Gene expression cancer rna-seq. UCI Machine Learning Repository, 2016. DOI: 10.24432/C5R88H.

P. Grassberger and I. Procaccia. Measuring the strangeness of strange attractors. *Physica D: Nonlinear Phenomena*, 9(1-2):189–208, 1983.

W. B. Johnson and J. Lindenstrauss. Extensions of lipschitz mappings into a hilbert space. *Contemporary Mathematics*, 26:189–206, 1984.

W. E. Johnson, C. Li, and A. Rabinovic. Adjusting batch effects in microarray expression data using empirical bayes methods. *Biostatistics*, 8(1):118–127, 2007.

B. Laurent and P. Massart. Adaptive estimation of a quadratic functional by model selection. *Annals of Statistics*, 28(5):1302–1338, 2000.

O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

O. Ledoit and M. Wolf. Numerical implementation of the quest function. *Computational Statistics & Data Analysis*, 115:199–223, 2017.

E. Levina and P. J. Bickel. Maximum likelihood estimation of intrinsic dimension. In *Advances in Neural Information Processing Systems*, volume 17, 2004.

V. A. Marchenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mat. Sb.*, 114(4):507–536, 1967.

L. McInnes, J. Healy, and J. Melville. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.

L. van der Maaten and G. Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9:2579–2605, 2008.

J. N. Weinstein et al. The cancer genome atlas pan-cancer analysis project. *Nature Genetics*, 45 (10):1113–1120, 2013.

A AI Usage

A.1 Tool and Scope of Use

All Python code, all figures, and the drafts of this report were produced with AI assistance. The primary tool was Claude Code (Anthropic), a command-line coding agent running Claude Sonnet 5 with read, write, and execute access to the project repository. A final review pass over this document was run on Claude Opus 5 (Section A, "Results Verification"). The Claude Code agent was the primary author of the implementation across all four project phases. The work was driven by written specifications (`project_framework/INSTRUCTIONS.md` and `SKELETON.md`), executed by the agent, then reviewed, tested, and in several documented cases corrected or rejected before being accepted. No module in `src/` was written without AI assistance, and Table 10 lists the specific files and functions involved. This appendix is a condensed record. The complete, incrementally maintained log, including every phase's full prompt list, is kept at `ai_usage_log.md` in the repository root.

A.2 What the Agent Generated

Table 10: Files and functions generated with AI assistance, by project phase. Phase 0 (environment, `requirements.txt`, `download_data.py`, `README.md`, and the initial `src/config.py`) was completed in an earlier session whose prompt transcript was not kept. The constants listed against `src/config.py` below are those added or revised in Phases 1–3.

Phase	File	Functions / content
1	`src/data_loading.py`	`load_expression_matrix()`, `load_labels()`, `validate_dataset()`
1	`src/preprocessing.py`	`log_transform()`, `drop_zero_variance_genes()`, `standardise()`, `filter_top_variable_genes()`
2	`src/synthetic.py`	`generate_gaussian_noise_baseline()`, `generate_low_dim_manifold_baseline()`
2	`src/intrinsic_dimension.py`	`correlation_integral()`, `default_radii()`, `_auto_detect_scaling_region()`, `estimate_correlation_dimension()`, `_kaiser_elbow()`, `_curvature_elbow()`, `pca_based_dimension()`, `knn_mle_dimension()`
2	`src/evaluation.py`	`build_intrinsic_dimension_summary_table()`, `reconcile_estimators()`
3	`src/embeddings.py`	`run_pca()`, `_pca_preprocess()`, `run_tsne()`, `run_umap()`
3	`src/random_projection.py`	`jl_target_dimension()`, `random_gaussian_projection()`, `evaluate_distance_preservation()`
3	`src/spectral_analysis.py`	`empirical_spectral_distribution()`, `marchenko_pastur_density()`, `marchenko_pastur_support()`, `identify_spectral_outliers()`
3	`src/geometry_diagnostics.py`	`pairwise_distance_concentration()`, `theoretical_concentration_bound()`
3	`src/anomaly_detection.py`	`shrinkage_covariance()`, `chi_squared_threshold()`, `mahalanobis_outliers()`
1–3	`src/viz.py`	`plot_class_balance()`, `plot_gene_mean_variance()`, `scree_plot()`, `log_log_correlation_plot()`, `scatter_embedding()`, `embedding_hyperparam_comparison_plot()`, `spectral_overlay_plot()`, `jl_distortion_plot()`, `distance_concentration_plot()`, `mahalanobis_distribution_plot()`
1–3	`src/config.py`	All analysis constants (`RANDOM_SEED`, `CORR_DIM_*`, `KNN_MLE_*`, `MANIFOLD_NOISE_STD`, `JL_EPSILON`, `SPECTRAL_N_TOP_GENES`, `MAHALANOBIS_ALPHA`, ...)
1–3	`run_pipeline.py`	Phase 1–3 orchestration, figure and summary-artifact export
1–4	`reports/phase_{1,2,3,4}.tex`	Per-phase methodology/results reports, consolidated into this document

A.3 Representative Prompts

The following are reproduced prompts from `ai_usage_log.md` in the project repository. They were chosen to span the four main phases and to give a good picture of how AI was used throughout. The prompts are quoted as logged in the file:

1. "Implement `src/preprocessing.py`: log1p transform, standardisation (zero mean, unit vari-

ance) applied *after* the log transform, and variance-based feature filtering retaining the top-k most variable genes."

2. "Implement the Grassberger–Procaccia correlation integral natively via `scipy.spatial.distance.pdist`, computing exact distances once and vectorising across a log-spaced radii array; auto-detect the linear scaling region via the plateau in the local log-log slope and justify the choice in the report."

3. "Implement the PCA-based estimator via the full eigenvalue spectrum, avoiding ever forming the (20531×20531) covariance matrix; report components needed for 90/95/99% variance and two elbow criteria (Kaiser, second-derivative curvature)."

4. "Implement the Levina–Bickel k-NN MLE estimator exactly per the given formula with a $(k-2)$ divisor, swept over $k = [5, 10, 20]$, with an epsilon guard against $\ln(1) = 0$ from distance ties under high-dimensional concentration."

5. "Implement synthetic baseline generators: isotropic Gaussian noise and a known-dimension linear subspace embedded in ambient space with additive noise, matched in shape to the real preprocessed data; run all three estimators on both baselines side by side with the real data in one summary table."

6. "Implement the Johnson–Lindenstrauss target-dimension formula and a random Gaussian projection, then empirically evaluate pairwise-distance distortion against the theoretical $(1 \pm \epsilon)$ bound."

7. "Implement the empirical spectral distribution of a tractable gene subset's correlation matrix and the theoretical Marchenko–Pastur density for overlay comparison, including the point-mass-at-zero case for aspect ratio > 1, and flag eigenvalues beyond the theoretical support edge as outliers."

8. "Implement pairwise distance concentration across feature-subsample sizes and a theoretical Hoeffding- or Chernoff-type bound for comparison, justified explicitly since Hoeffding itself requires bounded variables that standardised RNA-Seq features do not satisfy."

9. "Run the full pipeline, inspect every generated figure visually before writing the report, and cite only the real printed/computed numbers — no fabricated metrics."

A.4 Results Verification

Every number in this report was taken from a single pipeline run on the real dataset, and every figure was visually inspected before it was cited. Verification beyond inspection included an independent sanity check of the correlation-dimension implementation against a pure 5-D Gaussian control. This confirmed the estimator itself was correct, before the manifold-baseline anomaly was attributed to the noise parameter rather than a bug. It also included a cross-check of the empirical near-zero eigenvalue fraction (60.00%) against the independently derived theoretical point-mass prediction $1 - 1/\gamma = 60.00\%$ before that agreement was reported as a finding.

A subsequent review pass over the consolidated draft, run through the same agent on Claude Opus 5, audited every reported figure against the code that produced it and found four defects that the per-phase reports had carried forward. The curvature elbow criterion was implemented as the maximum of the *negated* second derivative, which inverted its sign. The corrected version recovers the ground-truth dimension 5 on the manifold baseline, where the original returned 4.

The Mahalanobis threshold used $p = 2{,}000$ degrees of freedom for a statistic confined by rank to 800 dimensions, making the test unfailable. The derivation in Section 4 was added as a result. The synthetic baselines were not standardised while the real data was, so the calibration compared unlike preprocessing. And the spectral aspect ratio used n rather than $n - 1$, understating an agreement that is in fact exact. All four were fixed in `src/` and the pipeline was re-run. The tables here reflect the corrected run. The review also dropped several claims that inspection could not support. These are recorded in Section 6 rather than removed silently.